

ALISHMA BHARDWAJ

+91 7064209567 | alishma25@gmail.com | [LinkedIn](#) | Bengaluru, India

Software Development Engineer at Amazon building distributed backend systems, payments infrastructure, and AI-driven operations tooling, with end-to-end architecture ownership and technical mentorship experience.

Work Experience

Amazon

Feb 2024 to Present

Software Development Engineer

Bengaluru, India

- Built an AI agent, DORA, on **Amazon Bedrock AgentCore** that answers operational questions directly, querying **DynamoDB, CloudWatch Logs, AWS Glue, SQS** through a **Model Context Protocol** tool layer, reducing manual investigation time from 30 minutes to under 5 minutes.
- Extended it into a self-service tool now used by 4 teams for dev testing, debugging & QA testing, adding log search, partial and full Glue job redrive for test runs, automated SQS redrive, and metrics views.
- Built a catalog ingestion pipeline (**AWS Glue, SQS, Lambda, SNS**) with a team of 3, replacing manual ingestion for in-store point-of-sale systems. Added idempotent processing, event ordering, and dead-letter recovery to safely handle duplicate and out-of-order events. Currently live in 10+ stores and designed for 900K records/day.
- Led the expansion of PayCheckout, the unified payment page for merchants on Amazon Pay India, into Mexico and Brazil, working cross-functionally with product, business, merchant, and engineering teams while technically leading 2 junior engineers through design and delivery. Owned the architecture and built a separate regional backend of **Java microservices** so each region could scale and function independently of the India stack.
- Built a payment recommendation system using customers' payment history across multiple **Amazon Pay India** merchants, increasing click-to-conversion on the recommended method to 17%.
- Built the full-stack rewards experience for PayCheckout, moving reward computation off the synchronous checkout path into an asynchronous orchestration flow and building the UI widget, backend reward fetching and calculation logic, and automatic application of the best available reward across merchants.
- Designed and built an outage simulation feature for the payment plugins layer, injecting **10+ exception scenarios across 12 checkout plugins** to prove that the fallback payment path let customers still place orders during a payment outage.

Ecolab Digital Center

Jul 2023 to Jan 2024

Software Associate Trainee

Bengaluru, India

- Automated an enterprise QA and release validation workflow using Power Automate and UiPath across 3 Azure-hosted applications, eliminating manual regression cycles and saving 8+ hours/week of operations effort.

Amazon

Jan 2023 to Jun 2023

Software Development Engineer Intern

Bengaluru, India

- Built a web-based payment configuration management application using Angular and REST APIs, integrating 5+ backend endpoints on an internal RPC/service framework and adding unit and integration test coverage with Jasmine, Karma, and Mockito to improve release reliability.

Education

Vellore Institute of Technology

2019 to 2023

Bachelor of Technology in Computer Science and Engineering; CGPA: 9.5/10

Chennai, India

Technical Skills

Languages: Java, Python, TypeScript, JavaScript, SQL

Backend & Architecture: Distributed Systems, System Design, Microservices, Event-Driven Architecture, Asynchronous Processing, System Integration, Message Queues, REST APIs, Java Microservices, FastAPI

AI & Agents: Amazon Bedrock AgentCore, Strands Agents SDK, Model Context Protocol (MCP), RAG, Vector Databases

Cloud & Data: Lambda, ECS/Fargate, Docker, SQS, SNS, DynamoDB (NoSQL), Glue, API Gateway, CDK, S3, CloudFront, CloudWatch, Elasticsearch/OpenSearch

DevOps & Frontend: CI/CD, CDK Pipelines, Infrastructure as Code, Unit & Integration Testing, Fault Injection & Resiliency Testing, Chaos Testing, Load Testing, HTML, React, Angular, Git, Agile/Scrum

Achievements

- **Amazon Spot Awards:** Received organization-level awards for Payment Resiliency Framework and PayCheckout International Expansion.